BIRMINGHAM CITY University

CMP6200/DIG6200
Individual Undergraduate Project (FYP)
Final Progress Review
Sign Language Translation System

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Date: 29/03/2026

1 Progress to Date - Summary

The project is currently in the **hardware prototyping and data training phase**. Since this is the first prototype, further development is planned, including an improved hardware iteration by the end of April, alongside continued work on data training and application development.

2 Artefact Design and Development

2.1 Implementation and Current State

The project has successfully transitioned from the research and design phase into a functional implementation. The current prototype consists of a hardware acquisition layer and a high-fidelity mobile interface.

Progressive Phases

The initial stages of the project have been successfully completed in accordance with the Gantt chart:

- Research & Design
- Software Foundation: Basic firmware has been developed to read analog signals, while the Android application "SignLink" has been built using a modern UI framework to handle future data visualization and translation (roughly seen below).

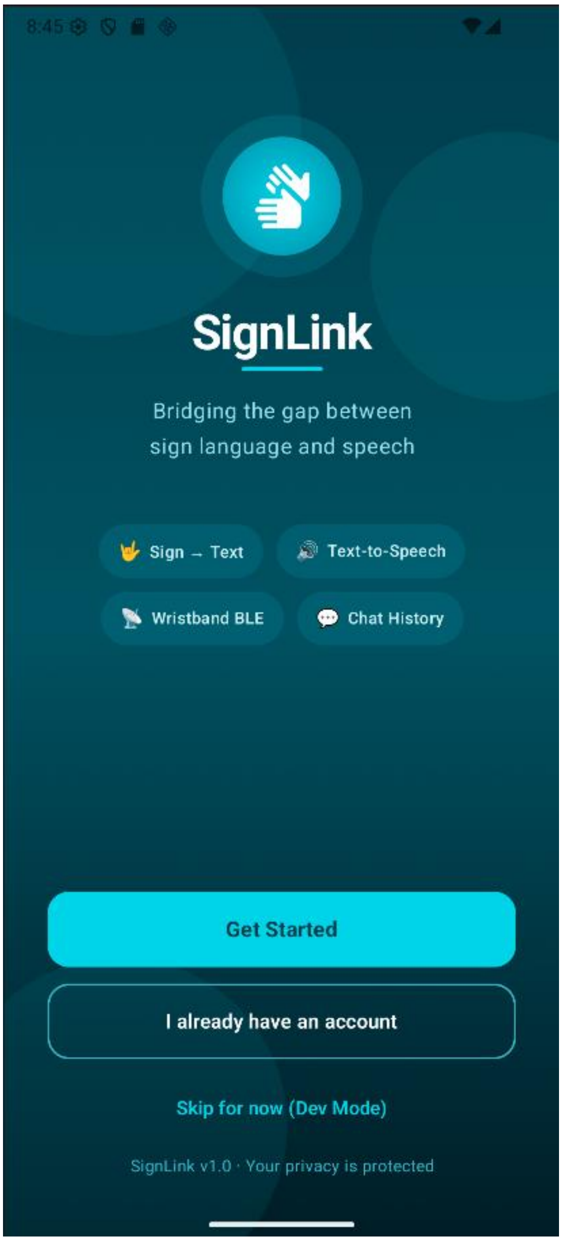


Figure 1. Start Screen

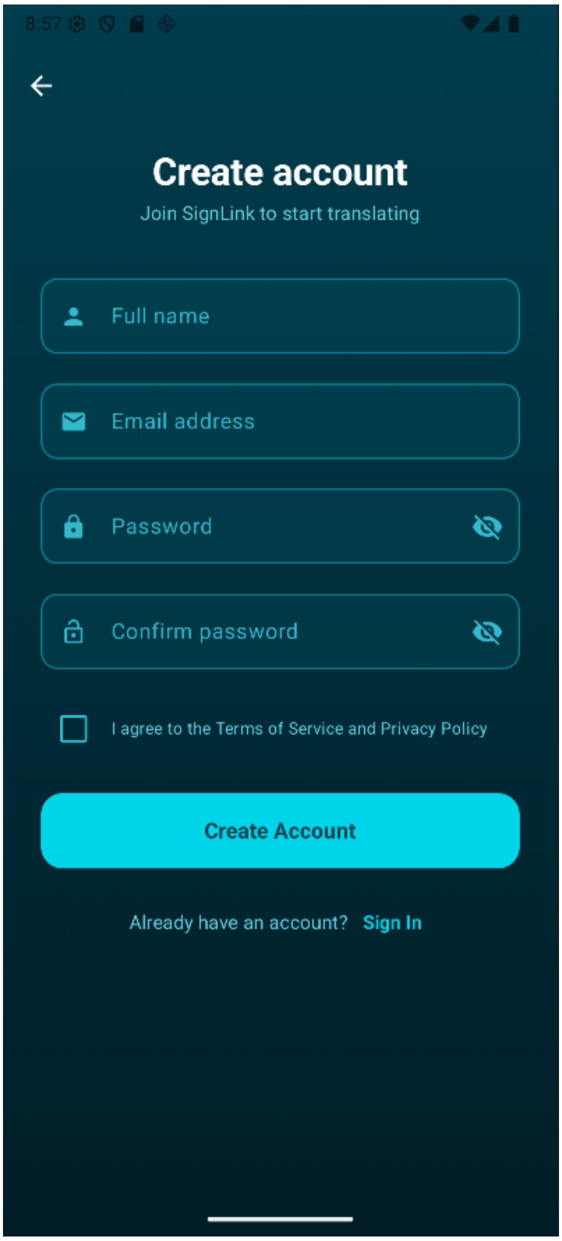
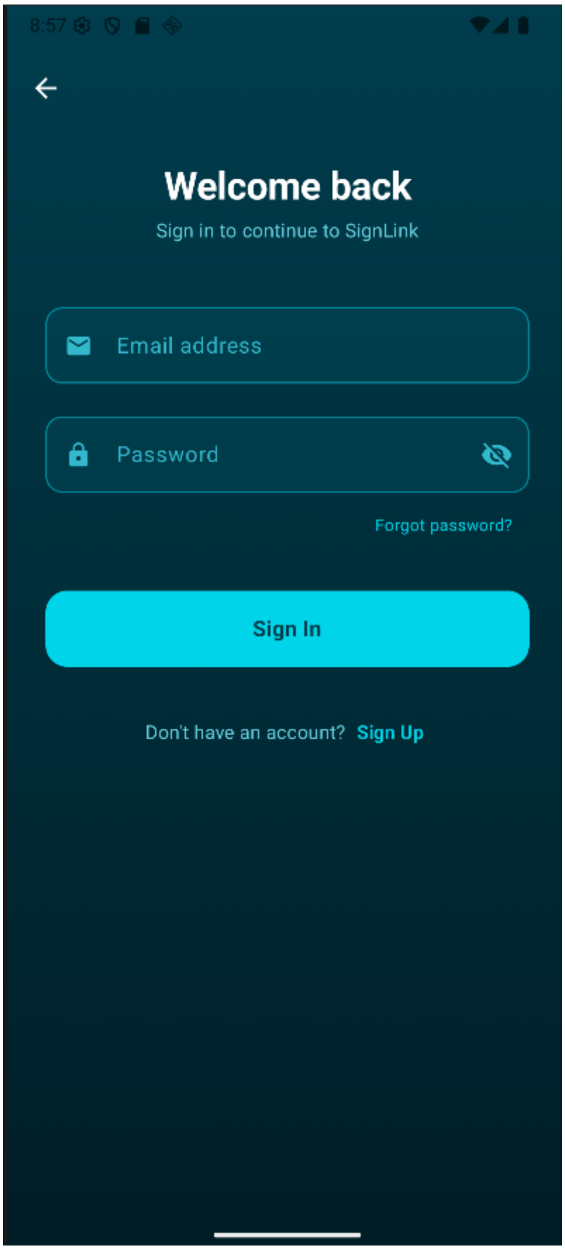


Figure 2. Sign-Up Screen



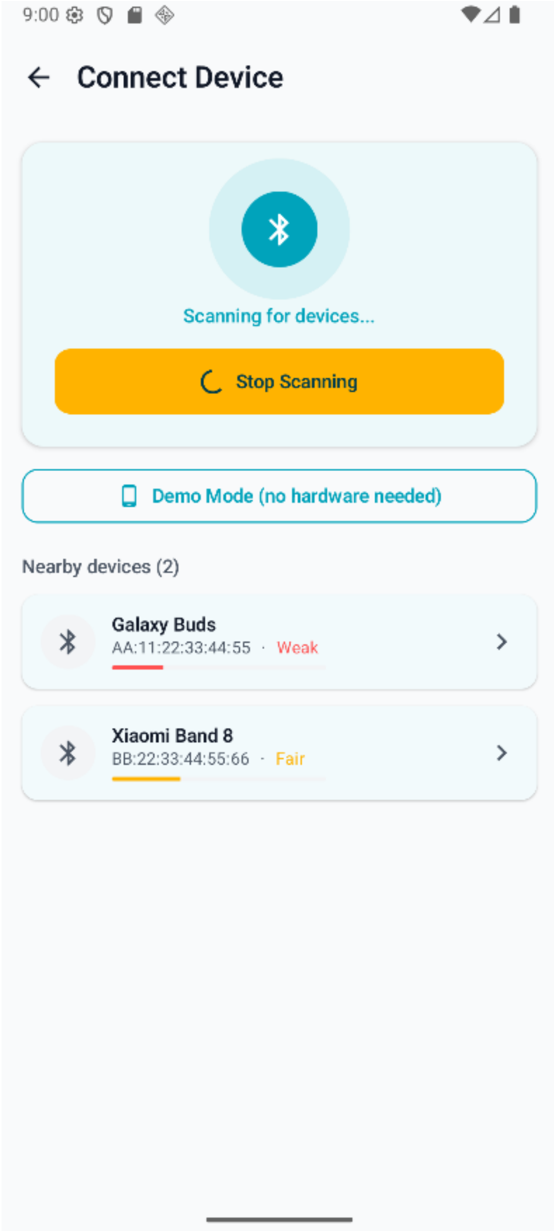
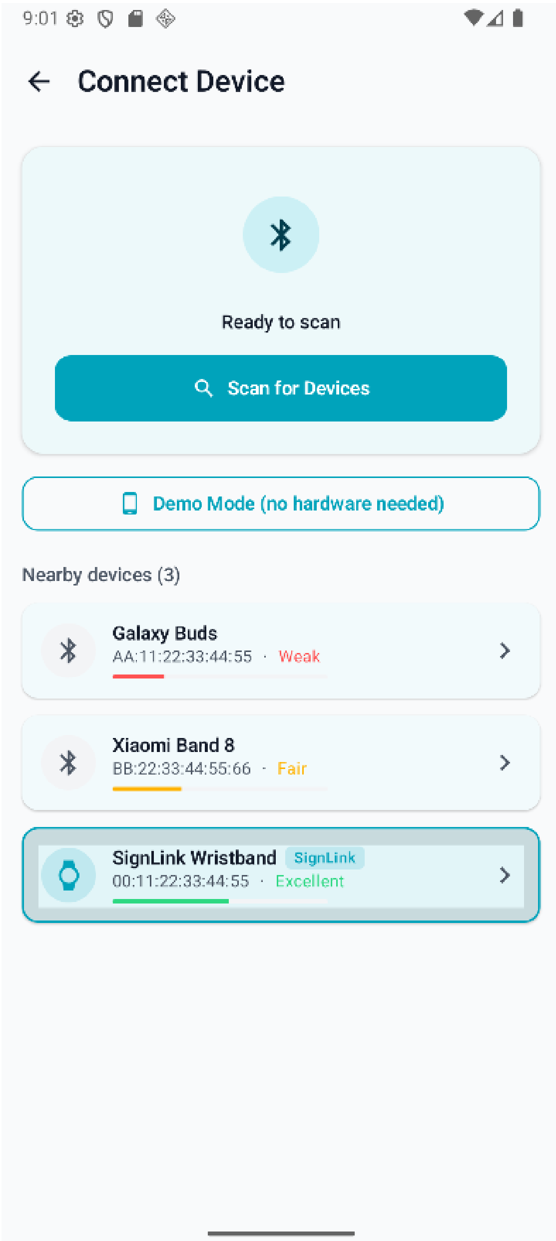


Figure 3. Sign-In Screen

Figure 4. Device Connection Screen



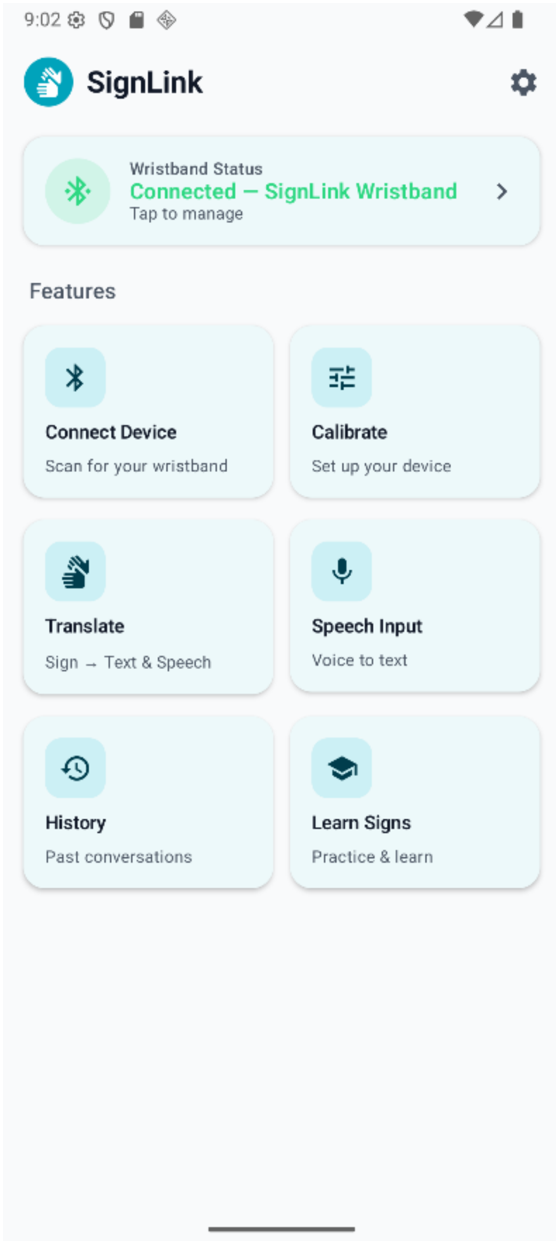
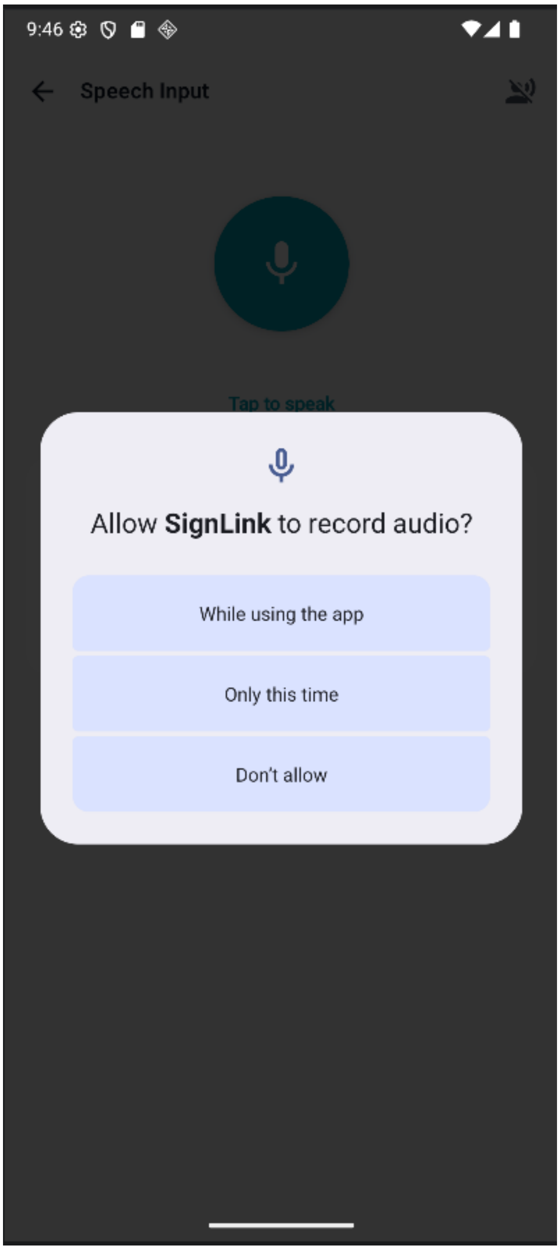


Figure 5. Devices Spotted (Demo)

Figure 6. Home Screen



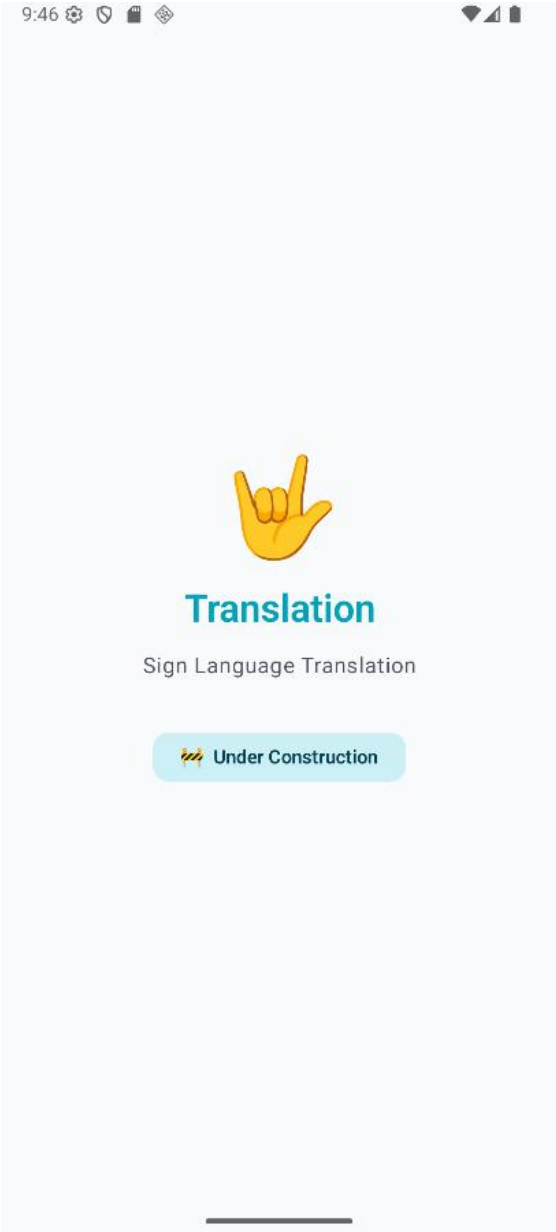


Figure 7. Voice to Text Screen (Feature)

Figure 8. SignLink's Translation Screen

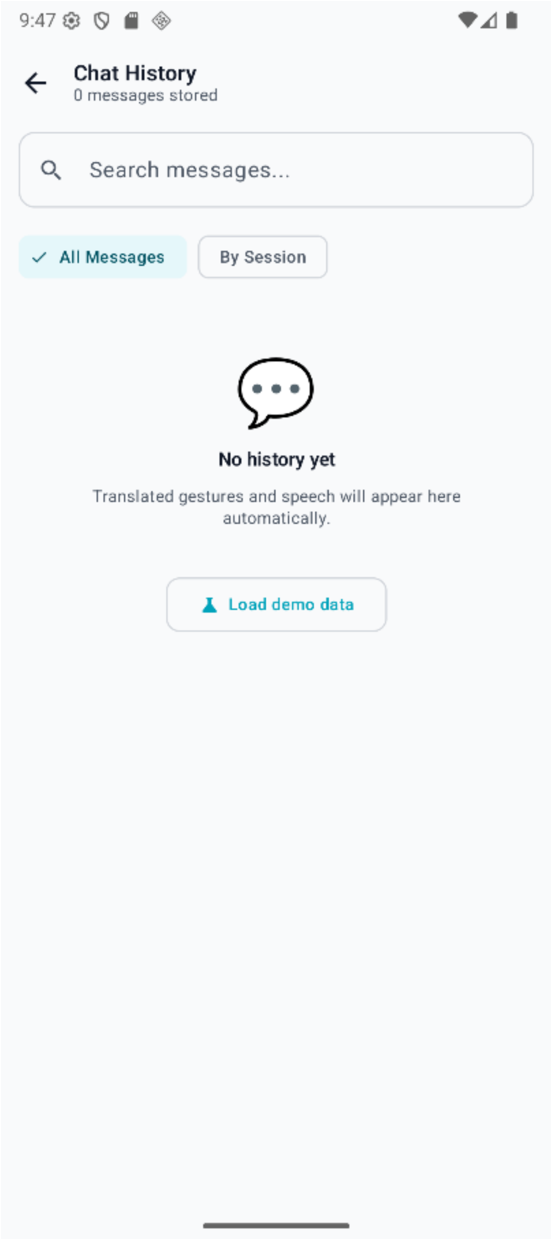


Figure 9. Chat History Screen

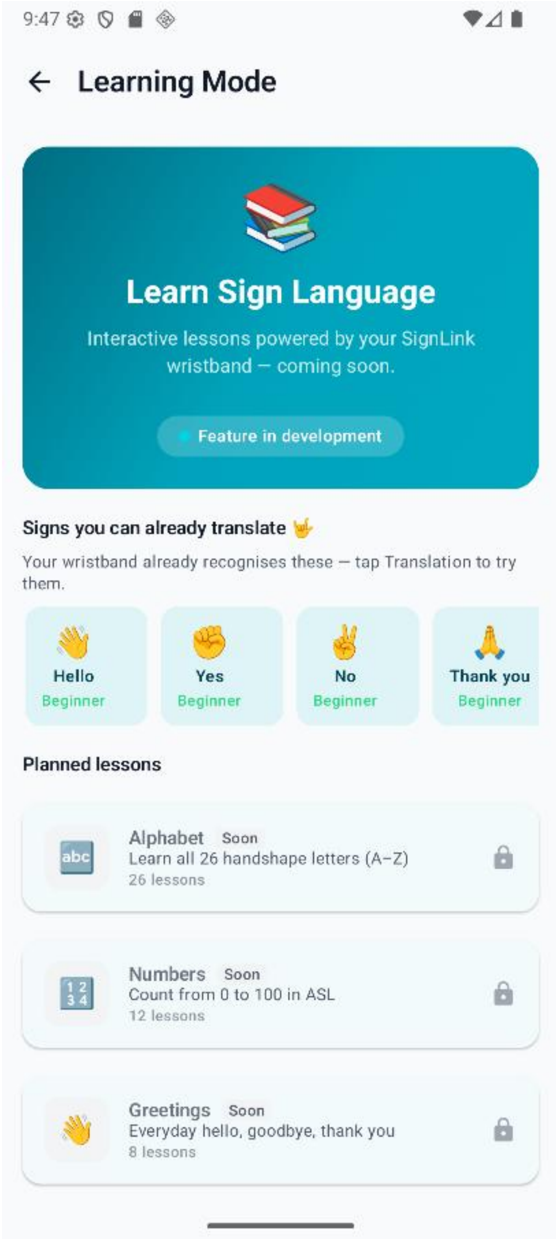


Figure 10. Learn ASL Screen

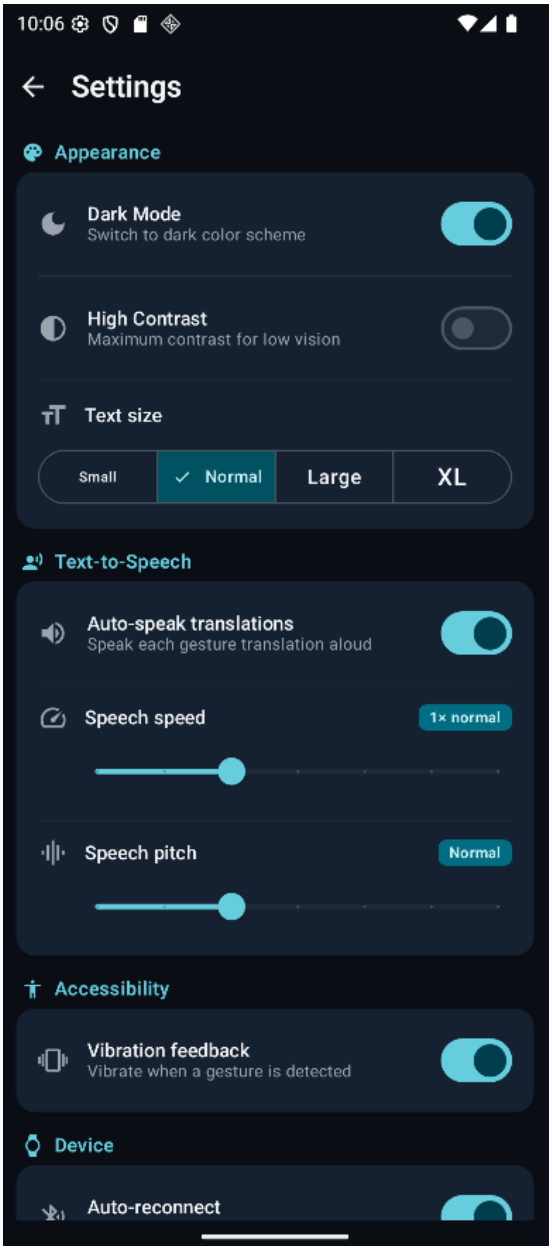


Figure 11. Settings Screen (Dark Mode has been toggled on resulting in visible color changes)

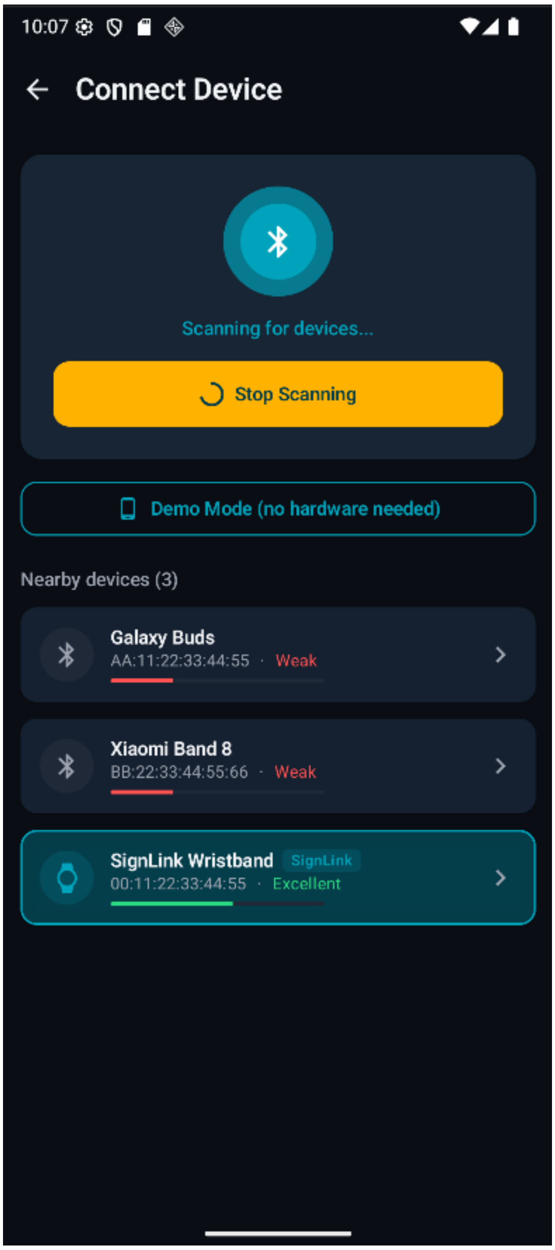


Figure 12. Device Connection Screen (Dark Mode)

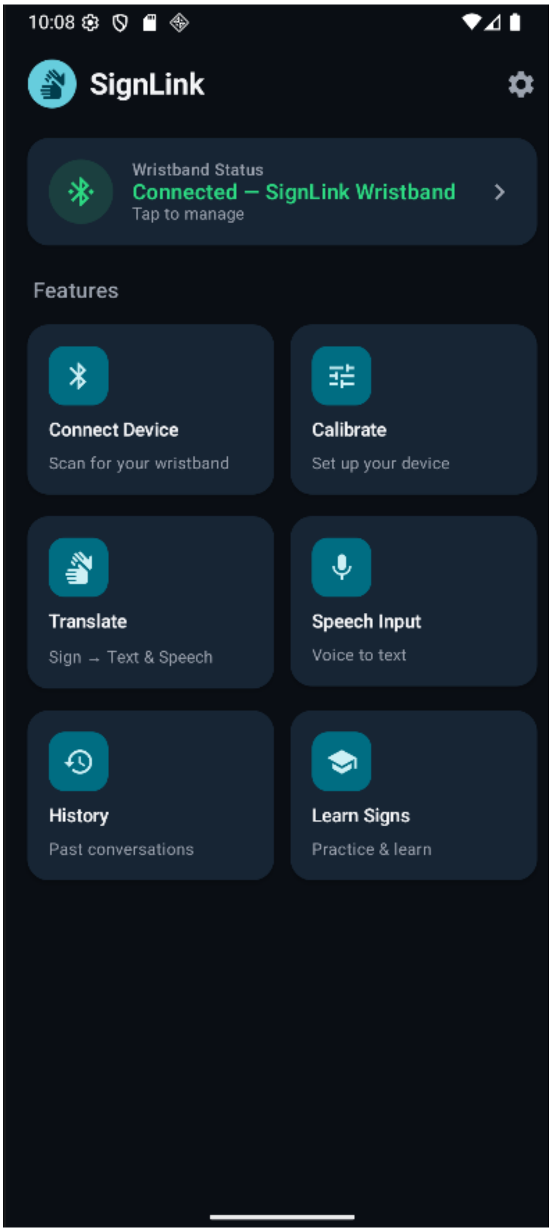


Figure 13. Home Screen (Dark Mode)

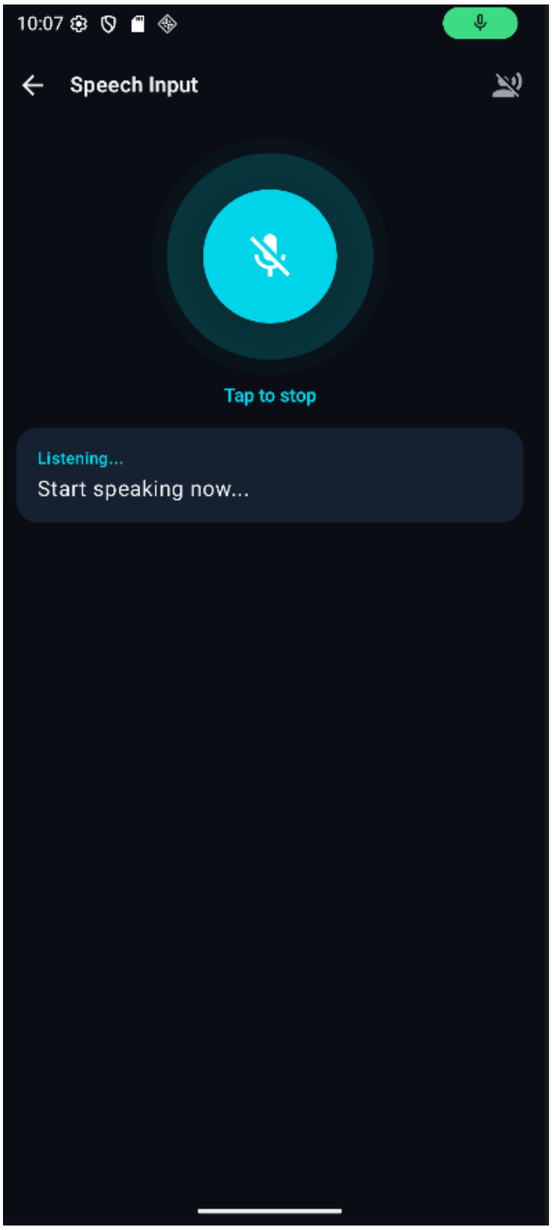


Figure 14. Voice to Text Screen (Dark Mode)

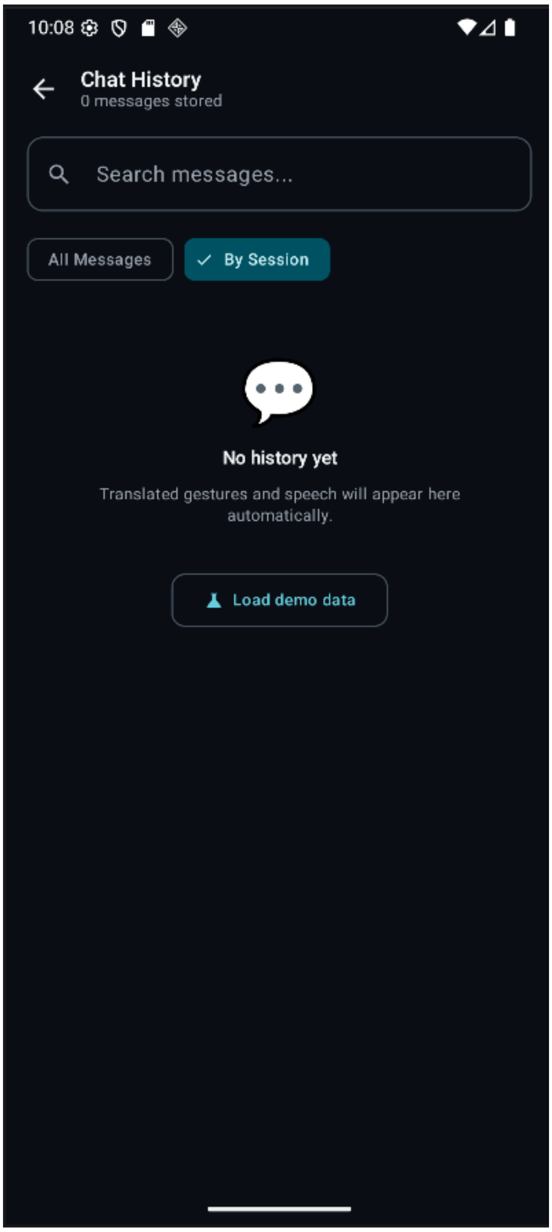


Figure 15. Chat History Screen (Dark Mode)

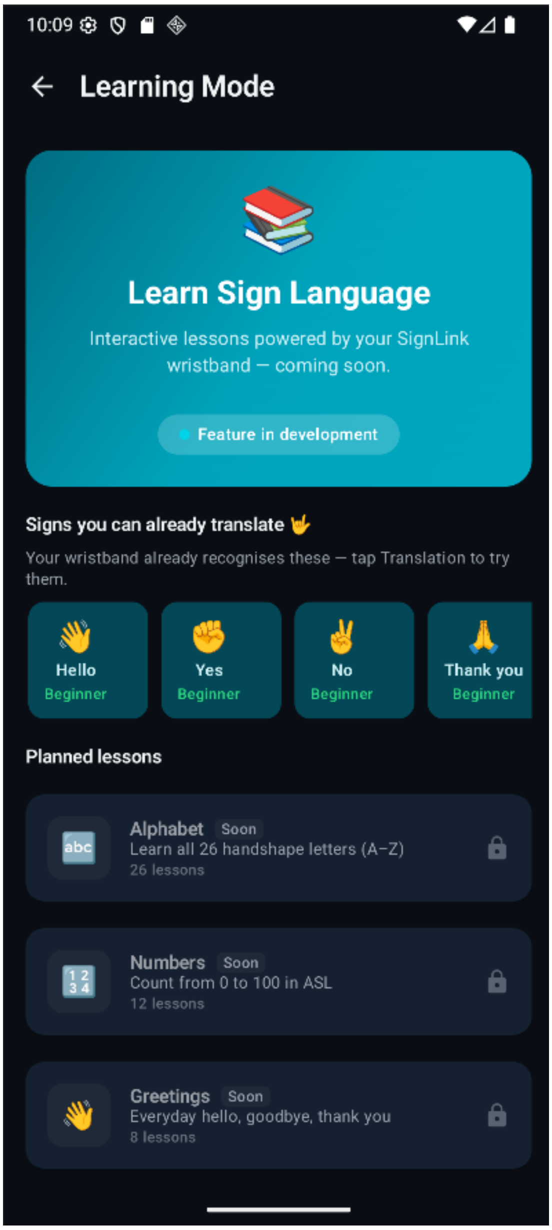
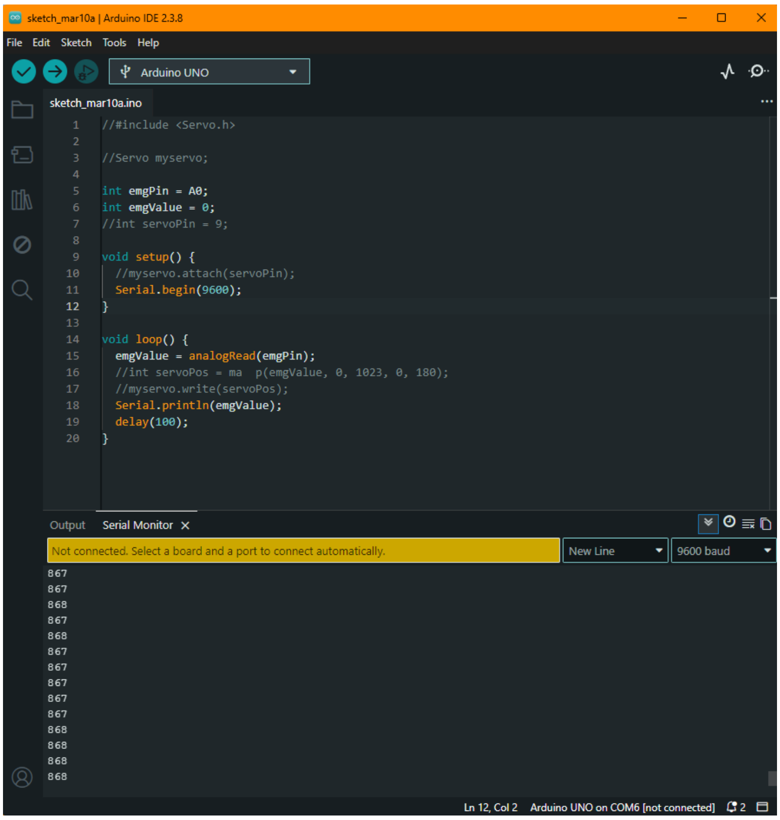


Figure 16. Learn ASL Screen (Dark Mode)

- Hardware Integration: The EMG sensor module is integrated with an Arduino Uno. Initial testing focused on power stability and grounding, which were identified as critical factors for signal integrity.



This is the code to view signals. This code does **not** specify what each high and low indicates.

Figure 17. Arduino Code to View Signals

Figure 18. Signals from EMG

The EMG sensor module has been successfully integrated with the Arduino Uno using appropriate jumper wire configurations.

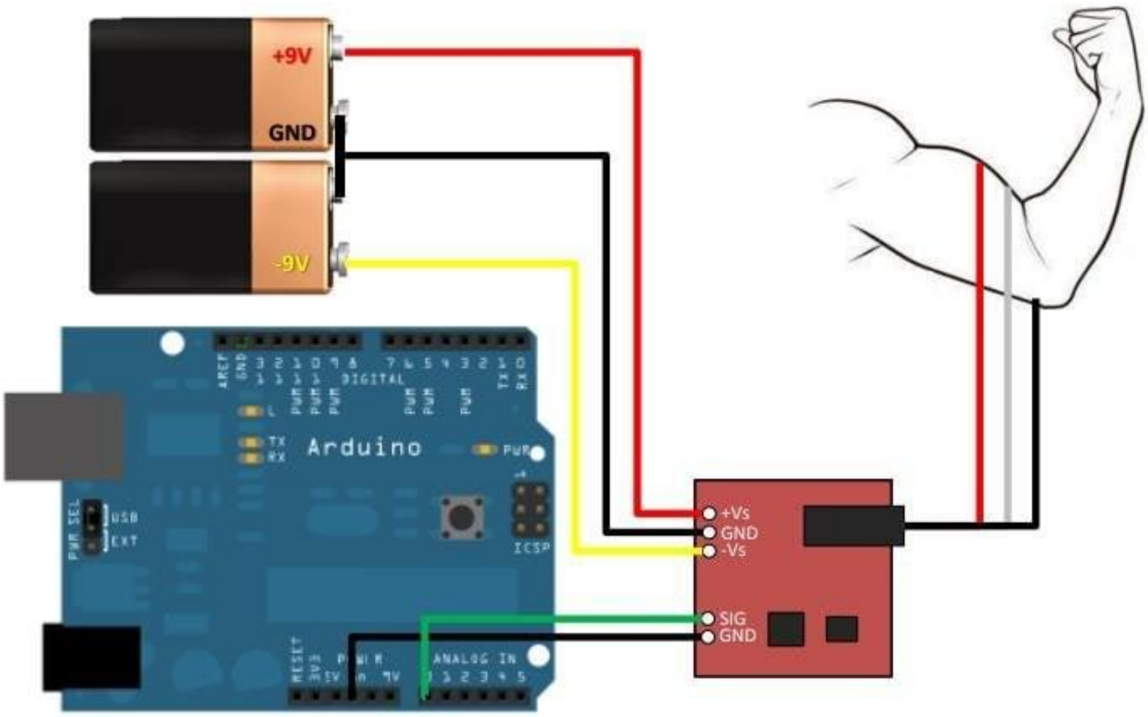


Figure 19. Circuit Diagram of the Current Setup

2.3 Analysis and Reflection

Through hands-on experimentation, several key insights emerged, altering the project's technical direction:

- **Hardware Sensitivity and Signal Integrity:** Initial tests confirmed the detectability of muscle activity, yet the raw electromyography (EMG) signals exhibited significant inconsistency due to noise and electrode sensitivity.
- **Technical Adaptation and Systematic Troubleshooting:** A critical turning point in development arose when unstable signal readings, initially thought to be caused by software bugs, were traced back to grounding issues in the hardware. This experience underscored the necessity of iterative testing and the importance of maintaining a controlled environment for bio-signal acquisition.
- **Validation of System Feasibility:** While the inconsistency of raw data indicates it is inadequate for direct classification, the successful capture of pertinent biological signals supports the project's fundamental premise.
- **Impact on Project Direction:** The advancements made so far have effectively established the system's input layer. However, the emphasis has now shifted from rapid gesture expansion to ensuring high-quality signal acquisition.

3 Updated Planning for Project Completion

To ensure the successful completion of the project within the given timeline, these tasks have been identified:

3.1 High-Priority Tasks

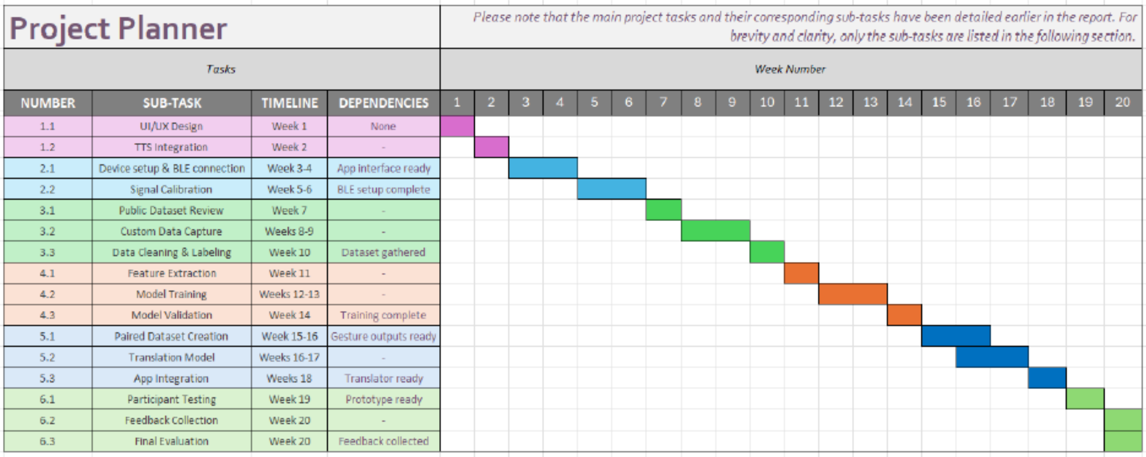
- **Refinement of Hardware Prototype**
  Improve electrode placement, grounding stability, and overall circuit reliability to ensure consistent EMG signal acquisition. This is prioritized as signal quality directly affects all subsequent stages of the system.

- **Signal Processing and Preprocessing**
  Implement filtering techniques (e.g., noise reduction and smoothing) and calibration methods to convert raw EMG signals into usable data. This step is essential before any classification or AI integration can be performed.

- **Data Collection and Training Dataset Development**
  Collect structured EMG data for different hand gestures to build a reliable dataset. This dataset will be used to train and evaluate machine learning models for gesture recognition.

- **Mobile Application Integration (Android)**
  Integrate the hardware system with the Android application using Bluetooth Low Energy (BLE) for wireless data transmission. Extend the application to display interpreted gestures as readable output.

Attached below is the Gantt chart provided for this project:



NUMBER	SUB-TASK	TIMELINE	DEPENDENCIES
1.1	UI/UX Design	Week 1	None
1.2	TTS Integration	Week 2	-
2.1	Device setup & BLE connection	Week 3-4	App interface ready
2.2	Signal Calibration	Week 5-6	BLE setup complete
3.1	Public Dataset Review	Week 7	-
3.2	Custom Data Capture	Weeks 8-9	-
3.3	Data Cleaning & Labeling	Week 10	Dataset gathered
4.1	Feature Extraction	Week 11	-
4.2	Model Training	Weeks 12-13	-
4.3	Model Validation	Week 14	Training complete
5.1	Paired Dataset Creation	Week 15-16	Gesture outputs ready
5.2	Translation Model	Weeks 16-17	-
5.3	App Integration	Weeks 18	Translator ready
6.1	Participant Testing	Week 19	Prototype ready
6.2	Feedback Collection	Week 20	-
6.3	Final Evaluation	Week 20	Feedback collected

Figure 20. Initial Gantt chart

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Attached further below is a rough extrapolation of the initial Gantt chart:

TASK ID	TASK TITLE	START DATE	DUE DATE	DURATION IN DAYS
STP1				
1	App Development	2-Mar-26	8-Jun-26	99
2	Write the Final Progress Review	23-Mar-26	27-Mar-26	5
	Review with advisor	1-Apr-26		
3	Setup of Hardware	25-Mar-26	27-Mar-26	3
4	Test Raw Signal Output	28-Mar-26	2-Apr-26	6
5	Fix issues and Validate Stable Readings	28-Mar-26	2-Apr-26	6
6	Define gestures / inputs and Record	3-Apr-26	7-Apr-26	5
7	Store and Label Data	3-Apr-26	7-Apr-26	5
	Review with advisor	8-Apr-26		
8	Filtering and Normalizing Signals	8-Apr-26	14-Apr-26	7
9	Train Model (Classification Logic)	12-Apr-26	20-Apr-26	9
10	Test Accuracy	17-Apr-26	24-Apr-26	8
11	Connect AI to App	22-Apr-26	29-Apr-26	8
	Review with advisor	29-Apr-26		
12	Write segment 1 and 2 of report	30-Apr-26	3-May-26	4
13	Write segment 3 and 4 of report	4-May-26	7-May-26	4
14	Setup Wearable and test signals	6-May-26	11-May-26	6
15	Fix issues and Validate Stable Readings	10-May-26	15-May-26	6
16	Define gestures / inputs and Record	15-May-26	20-May-26	6
17	Store and Label Data	20-May-26	22-May-26	3
	Review with advisor	21-May-26		
18	Retrain Model with Clean Data	23-May-26	28-May-26	6
19	Final Testing & Accuracy	27-May-26	2-Jun-26	7

20	Bug Fixes + Stability	31-May-26	4-Jun-26	5
	Review with advisor	3-Jun-26		
21	Write segment 5, 6 and 7 of report	25-May-26	7-Jun-26	14

Figure 21. Tasks and Timeline

3.2 Contents of the Final Report

1. Introduction

- Background on sign language communication and accessibility challenges
- Problem statement and project objectives

2. Literature Review

- Overview of existing sign language translation technologies
- EMG-based gesture recognition systems
- Limitations of current approaches and identified research gap

3. System Architecture and Design

- Overall system architecture (hardware + software integration)
- Data flow and system workflow design

4. Implementation

- Hardware setup and circuit configuration
- EMG signal acquisition process
- Software development (Arduino + Android application)
- BLE communication setup

5. Signal Processing and Model Development

- Signal preprocessing techniques (filtering, calibration)
- Feature extraction methods
- Machine learning model design and training process

6. Results and Evaluation

- Experimental setup and testing methodology
- Performance metrics (accuracy, latency, reliability)

- Analysis of system outputs

7. Discussion

- Interpretation of results
- Comparison with initial expectations
- Limitations of the current system

8. Conclusion and Future Work

- Summary of achievements
- Potential improvements (e.g., improved sensors, expanded gesture set, real-time optimisation)